

TEJASH RAI

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT) <i>Bachelor of Technology in Computer Science and Engineering</i>	Bhubaneswar, Odisha, India <i>Aug. 2024 -- May 2028</i>
Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Database Management Systems, Computer Architecture, Systems Programming, Internet Technology	

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL
ML/DL Frameworks: TensorFlow, LSTMs, Transformers, Scikit-learn, Matplotlib
LLMs & Agentic AI: LLMs, RAG, MCP (Model Context Protocol), Ollama, Faster-Whisper, Prompt Engineering
MLOps & Deployment: Docker, FastAPI, GitHub Actions (CI/CD), Azure, REST APIs
Tools & Platforms: Git/GitHub, VS Code, Google Colab, Google Cloud Platform

PROJECTS

AI-Based Methane Gas Forecasting System <i>Python, LSTMs, Transformers, FastAPI, Docker</i>	March 2026
- Architected a hybrid Deep Learning model utilizing LSTMs and Transformers to predict methane concentration levels, achieving high accuracy in forecasting potential leakages from multi-variate time-series sensor data. - Engineered an end-to-end MLOps pipeline with TensorFlow for model training, and containerized the application with Docker to serve real-time inference via FastAPI, ensuring sub-second safety alerts. - Developed a comprehensive monitoring dashboard, enabling stakeholders to visualize historical trends and receive automated push notifications for predicted anomalies.	
End-to-End MLOps CI/CD Pipeline <i>Python, Docker, GitHub Actions, Azure</i>	April 2026
- Engineered an automated MLOps CI/CD pipeline using GitHub Actions to manage machine learning model lifecycles and streamline continuous deployment. - Implemented automated data quality reports and model validation checks to monitor training architecture and handle neural network gradient degradation. - Bridged the gap between deep learning model training and robust production environments by utilizing infrastructure as code techniques.	
Jarvis – Agentic OS Copilot <i>Python, LLMs, MCP, Ollama, Faster-Whisper, RAG</i>	2026
- Architected an autonomous, multi-modal AI agent utilizing Qwen 2.5 via Ollama and Groq for high-speed LLM inference, orchestrating complex tool-calling and RAG workflows. - Developed an end-to-end speech-to-speech pipeline by integrating Faster-Whisper for low-latency audio transcription and Kokoro TTS for natural voice synthesis. - Leveraged the Model Context Protocol (MCP) to dynamically connect the LLM brain to external enterprise environments (Gmail, Calendar) while maintaining a secure local filesystem sandbox for production-grade reliability.	

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

E-Labs, KIIT University -- Senior Member	2025 -- Present
Sandrover Project (Kshitij, IIT Kharagpur): Led Team TRRAP in designing a 4WD rover, custom-calibrating a dual-ESC powertrain to reach the national finals. IOTRONIX MVP: Named “Most Valuable Player” for demonstrating technical excellence in hardware-software integration at the 2025 E-Labs flagship event. Joffee 3.0 Star Performer: Recognized as the top performer among peers for proficiency in Java, Data Structures, and industry-standard coding practices.	